Pythagorean Arithmetic Structure in the Minimal Pati–Salam Bracket Algebra

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Abstract

We record a structural observation about the dimensionless quantities that arise from the bracket $[e, \omega]$ in $\mathfrak{sl}(4, \mathbb{R})$ underlying the minimal Pati–Salam (PS) embedding. All native ratios produced by the construction — charge ratios, adjoint eigenvalues, the order-three saturation coefficient, and structural multiplicities — lie in the multiplicative subgroup $\langle 2, 3 \rangle$ of $\mathbb{Q}^\times$ generated by the primes 2 and 3. This is the arithmetic lattice of Pythagorean tuning. The lattice property is specific to the minimal PS embedding: the standard SU(5) and SO(10) constructions introduce factors of 5 through the 3+2 trace split and the rank-5 weight space respectively, and therefore do not preserve this structure. We test whether the broken-phase Standard Model mass and mixing spectrum retains any imprint of the lattice and find no statistically significant preference: the structure is an ultraviolet algebraic artifact, not a low-energy phenomenological prediction. The Koide ratio $\tilde{h}/h = 2/3$ is the single $\langle 2, 3 \rangle$ quantity that does survive into the broken phase, and only because it is a derived algebraic relation preserved by tree-level matching. We document the lattice observation, contrast it with SU(5) and SO(10), report the empirical negative result, and explicitly disclaim any further phenomenological consequence. The note is intended as a structural record, not a derivation of physics from music or vice versa.

1 Statement

Let $T_{B\text{–}L} = \mathrm{diag}(1/3, 1/3, 1/3, -1) \in \mathfrak{sl}(4, \mathbb{R})$ be the $B - L$ generator of the minimal Pati–Salam embedding, normalized so that the trace of the generator over the fundamental representation vanishes. The dimensionless quantities that arise mechanically from the bracket structure on $\mathfrak{sl}(4, \mathbb{R})$ with this generator are enumerated in Table 1.

Observation. Every entry in Table 1 lies in the multiplicative subgroup $\langle 2, 3 \rangle \subset \mathbb{Q}^\times$ generated by the primes 2 and 3. Equivalently, each native ratio is of the form $2^a \cdot 3^b$ for some $a, b \in \mathbb{Z}$.

This is mechanical. The 1/3 charge arises from $1/N_{\text{color}}$ with $N_{\text{color}} = 3$. The 4/3 gap is $1 - (-1/3) = 4/3 = 2^2 \cdot 3^{-1}$. The 16/9 saturation coefficient is the square of the gap. The multiplicities 3 and 9 are dimensions of color and color-squared structures. No prime greater than 3 enters the construction.

Remark (Connection to Pythagorean tuning). Pythagorean tuning is the system in which musical intervals are defined as the ratios in $\langle 2, 3 \rangle$, with 2 representing octave equivalence and 3 the lowest nontrivial harmonic [1]. Reduced into one octave by powers of two, the six native PS ratios of

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Quantity	Value	Prime structure
$T_{B–L}$ quark charge	1/3	3^{-1}
$\lvert T_{B–L}\rvert$ lepton charge	1	1
Eigenvalue gap of $T_{B–L}$	4/3	$2^2 \cdot 3^{-1}$
Nontrivial eigenvalue of $\mathrm{ad}_{T_{B–L}}$	4/3	$2^2 \cdot 3^{-1}$
Coefficient in $\mathrm{ad}^3_{T_{B–L}} = c\ \mathrm{ad}_{T_{B–L}}$	16/9	$2^4 \cdot 3^{-2}$
Color multiplicity	3	3^1
Dimension of zero eigenspace of $\mathrm{ad}_{T_{B–L}}$	9	3^2
BCH–Jacobi generation count	3	3^1

Table 1: Native dimensionless quantities of the minimal Pati–Salam bracket algebra and their prime factorizations.

Table 1 (above unison) correspond to the unison (1), the major second $9/8 \leftrightarrow 9$ reduced, the perfect fourth $4/3 \leftrightarrow 1/3$ or $4/3$ reduced, the perfect fifth $3/2 \leftrightarrow 3$ reduced, and the Pythagorean minor seventh 16/9.

This is not a derivation of music from physics, nor of physics from music. It is an identification of two systems that happen to live on the same arithmetic lattice for different mechanistic reasons. The PS lattice is generated by $N_{\mathrm{color}} = 3$ and the charge difference $|q_q - q_\ell| = 4/3$ (factoring as $2^2 \cdot 3^{-1}$); the Pythagorean lattice is generated by octave doubling and the third harmonic.

2 Contrast with SU(5) and SO(10)

The arithmetic lattice property of Observation 1 is *not* a generic feature of grand-unified embeddings. We demonstrate this for the standard alternatives.

SU(5). In the Georgi–Glashow embedding, the hypercharge generator on the fundamental **5** is $Y \propto \mathrm{diag}(-2/3, -2/3, -2/3, +1, +1)$. The trace constraint $3 \cdot (-2/3) + 2 \cdot (+1) = 0$ forces the $3+2$ split that defines $\mathrm{SU}(3) \times \mathrm{SU}(2) \subset \mathrm{SU}(5)$. The eigenvalue gap is $1 - (-2/3) = 5/3$, which factors as $5 \cdot 3^{-1}$. The order-three saturation coefficient (where it exists) is $(5/3)^2 = 25/9 \in \langle 3, 5\rangle$. The fundamental representation has dimension 5.

SO(10). The SO(10) construction has rank 5 and contains the spinor representation of dimension $16 = 2^4$. The rank-5 Cartan structure introduces 5 as a structural prime; the embedding $\mathrm{SO}(10) \supset \mathrm{SU}(5) \times U(1)$ inherits the 5/3 gap of SU(5).

Construction	Native lattice	Reason for primes
Minimal Pati–Salam ($\mathfrak{sl}(4,\mathbb{R})$, $T_{B–L}$)	$\langle 2, 3\rangle$	$N_{\mathrm{color}} = 3$, charge gap $= 4/3$
SU(5) (Georgi–Glashow)	$\langle 2, 3, 5\rangle$	$3 + 2$ trace split, fundamental dim 5
SO(10) (spinor embedding)	$\langle 2, 3, 5\rangle$	rank 5, inherits SU(5) gap 5/3

Table 2: Native arithmetic lattices of three standard unification embeddings. Only the minimal Pati–Salam case lies entirely in $\langle 2, 3\rangle$.

Remark (Just intonation). Just intonation extends Pythagorean tuning by including the prime 5, giving intervals such as the major third 5/4 and the major sixth 5/3 [1, 2]. The arithmetic lattice of just intonation is $\langle 2, 3, 5\rangle$, which is precisely the lattice into which the SU(5) and SO(10) embeddings fall. The parallel between the unification chain PS $\subset$ SU(5) or SO(10) and the tuning chain Pythagorean $\subset$ just intonation is again purely arithmetic; we do not assign physical or musical significance to it.

3 What is and is not claimed

For clarity we state explicitly what this note does and does not claim.

What is claimed.

1. The native dimensionless ratios produced by the bracket structure on $\mathfrak{sl}(4, \mathbb{R})$ with $T_{B\text{–}L}$ are exactly the elements of $\langle 2, 3\rangle$ shown in Table 1.
2. This arithmetic lattice coincides with the lattice of Pythagorean tuning.
3. The lattice property is specific to the minimal PS embedding. SU(5) and SO(10) introduce the prime 5 for independent group- theoretic reasons (the $3 + 2$ trace split and the rank-5 Cartan structure respectively).

What is not claimed.

1. No statement is made that physics is musical, or that music is physical, or that either system derives from the other.
2. No phenomenological prediction follows from the lattice observation. We tested this directly in Section 4 and found that the broken-phase Standard Model spectrum does not retain a detectable imprint of the $\langle 2, 3\rangle$ lattice. The structural observation is confined to the unbroken algebraic sector.
3. No claim is made that one should prefer the minimal Pati–Salam embedding over SU(5) or SO(10) on the grounds of arithmetic simplicity. The lattice property is a feature, not an argument.
4. No claim of universality is made. Other gauge embeddings, other Lie-algebra constructions, and other physical systems may or may not exhibit similar lattice structure; we have not surveyed them.

4 Empirical test: does the broken-phase spectrum preserve the lattice?

The arithmetic-lattice property of Observation 1 holds in the unbroken algebraic sector. A natural empirical question is whether the broken-phase mass spectrum and mixing angles preserve any detectable imprint of this structure. We run the test here and report the result.

Method. For each measured dimensionless ratio r, define the lattice distance

$$\varepsilon(r;\, \mathcal{L}) \;=\; \min_{(a_1,\ldots,a_k)\in\mathbb{Z}^k} \Big|\, \log_2 r \;-\; (a_1 \log_2 p_1 + \cdots + a_k \log_2 p_k) \,\Big|, \tag{1}$$

where $\mathcal{L} = \langle p_1, \ldots, p_k\rangle$ is the multiplicative subgroup of $\mathbb{Q}^\times$ generated by the primes $p_1, \ldots, p_k$, and the search is over integers $|a_i| \le N$ for some fixed cap N. This is the minimum log-distance from r to any lattice point.

For each set of measured ratios, we compute the mean of ε and compare against a null distribution: 10^4 ratios drawn uniformly in log scale over the same range as the data. A genuine lattice signal would manifest as a significantly lower mean lattice distance for the data than for the null, and as a sharper improvement when the "correct" lattice is used.

Data. We use PDG 2024 central values: charged-lepton pole masses, $\overline{\mathrm{MS}}$ light-quark masses at 2 GeV, $\overline{\mathrm{MS}}$ c and b masses at their own scales, the top pole mass, and the CKM magnitudes from the global fit. We compute three categories of ratio: intra-generation (within a single doublet), inter-generation (Yukawa hierarchy), and CKM-derived.

Result. Table 3 reports the mean lattice distance against $\langle 2,3\rangle$, $\langle 2,3,5\rangle$, and $\langle 2,3,5,7\rangle$ for each ratio category at $N=5$, alongside the corresponding null-distribution means.

	$\langle 2,3\rangle$		$\langle 2,3,5\rangle$		$\langle 2,3,5,7\rangle$	
Ratio category	Data	Null	Data	Null	Data	Null
Intra-generation	0.032	0.040	0.008	0.007	0.001	0.002
Inter-generation	0.080	0.065	0.006	0.007	0.001	0.002
CKM-derived	0.035	0.049	—	—	—	—

Table 3: Mean lattice distance $\langle\varepsilon\rangle$ in $\log_2$ units for measured Standard Model ratios versus null distribution (uniform random in same log range). All z-scores fall in $[-0.7,+0.3]$; no category shows a statistically significant signal.

The data sits at 0.6–1.2× the null mean across all categories and all lattices. The largest absolute z-score is -0.71 for CKM ratios on $\langle 2,3\rangle$ at $N=7$, which would be significant only at the $\sim$0.24 level (one-tailed).

Proposition 1 (Negative result: no detectable lattice imprint in the broken phase)**.** *The dimensionless mass and mixing ratios of the broken-phase Standard Model do not exhibit a statistically significant preference for the multiplicative subgroup $\langle 2,3\rangle$ of $\mathbb{Q}^\times$. The discriminating diagnostic — whether adding the prime 5 to the lattice improves data fit more than it improves the null — shows no asymmetry: adding primes reduces lattice distance equally for the data and the null, consistent with the data being lattice-generic.*

Interpretation. The negative result is itself the clean conclusion. The arithmetic-lattice structure of Observation 1 is confined to the algebraic sector: it is a property of the *representation theory* of the bracket $[e,\omega]$ in $\mathfrak{sl}(4,\mathbb{R})$, not of the broken-phase phenomenology. The arithmetic-lattice structure does not survive into the broken-phase observables after symmetry breaking, renormalization-group evolution, and threshold matching. We do not isolate which of these mechanisms is responsible for the loss of lattice structure; the empirical statement is that the imprint is not detectable in the infrared.

This places the observation firmly in the category of an *ultraviolet algebraic artifact*: a correct, mechanical fact about the construction that does not propagate measurably to infrared observables. The structural identification with Pythagorean tuning remains valid as a statement about the algebraic sector; it does not extend to a prediction about masses or mixing.

The Koide ratio $\tilde{h}/h = 2/3$ derived in [3] is the one place where a $\langle 2,3\rangle$ quantity does survive into the broken-phase Yukawa structure, but it is a single ratio at a fixed scale; it does not generalize to a lattice-wide structure across the full mass spectrum.

Acknowledgment

The observation reported here was identified through repeated adversarial compression: an initial broader claim about "cosmic musical structure" was tested against control distributions of small rationals, narrowed under the recognition that the relevant arithmetic constraint is membership in $\langle 2, 3 \rangle$ rather than statistical coincidence, and finally compressed to the present structural observation contrasted against SU(5) and SO(10). The final form is what survives that process.

References

[1] A. Barker, *Greek Musical Writings, Vol. II: Harmonic and Acoustic Theory*, Cambridge University Press, Cambridge (1989).

[2] W. A. Sethares, *Tuning, Timbre, Spectrum, Scale*, 2nd ed., Springer-Verlag, London (2004).

[3] B. Wallace, *Colour from Gravity: Pati–Salam Structure from the Palatini Bracket*, Preprint TR-2026-FF06a (2026).